

FitBuddy – AI Fitness Plan Generator

Project Documentation Phase – Page 1

Documentation Phase Overview

The Documentation Phase records the complete FitBuddy project in a clear and organized form so that its purpose, design, implementation and results can be understood.

- Project overview
- Requirements
- Design
- Implementation
- Testing
- Results

Documentation Purpose

Good documentation helps users, teachers and developers understand how the system works and how it can be maintained.

FitBuddy – AI Fitness Plan Generator

Project Documentation Phase – Page 2

Project Introduction

The document introduces FitBuddy as an AI-based fitness planning web application using Gemini services.

- Problem statement
- Objectives
- Scope
- Target users
- Expected benefits

System Overview

The system accepts fitness information, generates a personalized workout plan and nutrition tip, and supports feedback-based updating.

FitBuddy – AI Fitness Plan Generator

Project Documentation Phase – Page 3

Requirement Documentation

The requirements section records functional and non-functional requirements identified during analysis.

- User inputs
- Workout generation
- Nutrition tip
- Feedback update
- Database storage
- Admin/user view

Technology Documentation

The technologies and dependencies used in the project are listed with their purpose.

FitBuddy – AI Fitness Plan Generator

Project Documentation Phase – Page 4

Design Documentation

The design section explains architecture, modules, routes, interface, database and workflow.

- System architecture
- Module design
- UI design
- Route design
- Database design
- AI integration

Diagrams and Flow

Architecture diagrams, flowcharts and data-flow representations can be included to visually explain system operation.

FitBuddy – AI Fitness Plan Generator

Project Documentation Phase – Page 5

Implementation Documentation

The implementation section describes how the planned modules were converted into working code.

- AI generator
- Nutrition generator
- Database functions
- FastAPI routes
- Templates
- Configuration

Code Organization

Important files and their responsibilities should be documented so that the project structure is easy to understand.

FitBuddy – AI Fitness Plan Generator

Project Documentation Phase – Page 6

Testing Documentation

Testing records the test cases, expected results, actual results and corrections made during development.

- Unit testing
- Route testing
- AI testing
- Database testing
- Integration testing
- UI testing

Test Evidence

Screenshots of successful pages, generated plans and relevant outputs can be included as evidence.

FitBuddy – AI Fitness Plan Generator

Project Documentation Phase – Page 7

User Documentation

The user guide explains how to run FitBuddy and use the main features.

- Open application
- Enter details
- Generate plan
- Read nutrition tip
- Submit feedback
- View updated plan

Administrator Documentation

The administrative view and stored user/plan information should be explained where applicable.

FitBuddy – AI Fitness Plan Generator

Project Documentation Phase – Page 8

Deployment Documentation

The deployment section records environment setup, dependency installation, API configuration and server execution.

- Virtual environment
- Requirements installation
- Environment variables
- Uvicorn command
- Deployment checks

Troubleshooting

Common issues such as missing dependencies, API key errors, template errors and database problems should be recorded with their solutions.

FitBuddy – AI Fitness Plan Generator

Project Documentation Phase – Page 9

Results and Conclusion

The final documentation records the achieved functionality and summarizes the project outcome.

- Personalized plans
- Nutrition guidance
- Feedback updating
- Persistent storage
- Web interface

Future Enhancement Documentation

Possible future enhancements can include richer profiles, additional fitness goals, improved prompts and more advanced analytics.

FitBuddy – AI Fitness Plan Generator

Project Documentation Phase – Page 10

Documentation Summary

Complete documentation provides a permanent record of the FitBuddy project's requirements, design, development, testing and usage.

- Clear structure
- Screenshots and diagrams
- Test evidence
- Setup instructions
- Final results

Documentation Conclusion

A well-organized document supports project evaluation, demonstration and future maintenance.